ABIR DEOL

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EDUCATION

University of Alberta · Bachelor of Science, Computing Science *Expected 2028*

TECHNICAL SKILLS

Languages: C, C++20, Rust, Python, Go, TypeScript, JavaScript, GLSL, WGSL, SQL

Systems: Memory safety and sanitizers (ASan, UBSan, Miri), concurrency, memory management, distributed consensus, storage engines, GPU rendering and compute (Vulkan, WebGPU, SPIR-V), TCP and WebSockets

Tools: Linux, Git, GDB, CMake, Vulkan validation layers, LLVM/Clang, SDL2, FastAPI, React, SQLite

OPEN-SOURCE CONTRIBUTIONS

52 pull requests and 12 issues across 17 upstream projects, 34 merged. Most began as a bug I found, reported, and then fixed myself. Write-ups at abirdeol.tech/work.

Vulkan and SPIR-V · validation layers, SPIRV-Tools, SPIRV-Cross, glslang, Mesa, wgpu | *C++, SPIR-V, GPU-AV, RADV, lavapipe* `25 MERGED`

- Fixed a GPU-AV segfault in `vkCreateComputePipelines` on any `coopMatLoad` or `coopMatStore` with a runtime stride, which the instrumentation pass read as a constant (#13098 to #13104). Fixed an abort when GPU-AV decoded a shader offset that never named an instruction, found under a llama.cpp workload (#13134 to #13150, five review rounds). Mesh shading fixes in the validation layers (#13246, #13260), glslang (#4467), wgpu (#10496, #10497) and lavapipe (!44818).
- Ran the full suite on lavapipe, Mesa's CPU Vulkan driver, at the maintainer's suggestion: added validation that linked pipeline libraries with independent sets never had (#13208), made tests portable (#13209, #13217), fixed a false warning on depth/stencil resolves (#13241), moved export checks to allocation (#13245), and reported four RADV and Mesa driver bugs with the VUID showing the application was valid. Two became RADV fixes and CTS tickets.

ggml-org · llama.cpp and whisper.cpp | *C++, GLSL, Vulkan, CI* `5 MERGED`

- Found, with the validation layers, that both Vulkan `im2col` shaders declared a `buffer_reference` with no alignment, so every store was emitted as `Aligned 16` against a real 2 or 4 byte stride. One word per shader took 40 VUID violations to zero (#28960 to #28996).
- Fixed the whisper.cpp test suite aborting on every `GGML_BACKEND_DL=ON` build: no backend registers until something calls `ggml_backend_load_all()`, and the tests were the only callers that did not (#4030 to #4031). Documented the stream example's VAD timestamps and model-path failure mode (#4019).
- Fixed a heap out-of-bounds read reachable from `whisper-cli --vad` on any user-supplied VAD model, found under AddressSanitizer with a crafted 94-byte model (#4064), and added the CI leg that would have caught the break above (#4047).

AMD · lemonade SDK and ROCm TheRock | *C++, Python* `3 MERGED`

- AMD GPUs on Linux were named by their raw KFD `gfx_target_version`, so an RX 9070 XT reported itself as `120001`. That field also keys the ROCm architecture lookup, so the readable name had to go beside it rather than replace it (#3592 to #3601).
- Traced embedding requests over 512 tokens failing with a 500 under an 8192 token context: an embedding model is non-causal, so the physical micro-batch is the real ceiling and had regressed to the default (#3591 to #3611).

Memory-safety fixes | *Hugging Face candle and tokenizers, espeak-ng, pdf-rs, Rust cargo*

- Undefined behaviour loading quantized GGML tensors, a byte slice cast to blocks with no length or alignment check; and a heap overflow in `Tensor::from_raw_buffer` reachable from safe code, where a seven-byte f32 buffer wrote past its allocation and returned `Ok` (candle #3963, #3965, found with Miri).
- A malformed BPE merge list producing invalid UTF-8 through `from_utf8_unchecked`; the fix deletes the unsafe block (tokenizers #2398). An unsigned subtraction reading before a heap string on ordinary voice selection, found under UBSan (espeak-ng #2530).

PROJECTS

colossus · Virtualized Geometry Renderer | *C++20, Vulkan, GLSL, WebGPU*

- Built Nanite-style virtualized geometry from scratch: a builder splits scanned models into 128-triangle clusters and simplifies them into a crack-free LOD hierarchy with proven error bounds; the renderer streams pages from disk and picks, per cluster on the GPU, the coarsest level within a pixel.
- Two-pass occlusion culling, mesh shaders plus a compute rasterizer for pixel-sized triangles, and virtual shadow maps draw 900 scanned statues (15.9 billion triangles at full detail) at 1080p in 1.51 ms on an RX 9070 XT, and a million instances in 1.78 ms. Ported to WebGPU as a live demo.

oracle_of_delphi · Local Voice Assistant | *Rust, C++, TypeScript, Three.js*

- Built a fully local voice assistant as cooperating processes behind one executable: a real-time C++ audio layer for capture, VAD and barge-in, a Rust agent orchestrator, and a privileged actuator daemon. Boots offline with no GPU, model or credentials, then swaps in real backends behind traits.
- Isolated all OS actuation in the privileged daemon, treating the language model as an untrusted planner: it recomputes the required capability from the operation itself, gates irreversible actions behind confirmation, and audits over a `SO_PEERCRED` socket. 481 Rust tests and 933 C++ checks pass on Linux, macOS and Windows CI.

nexus_db · Embedded LSM Key-Value Store | *C++17, Python/FastAPI, React*

- Built a storage engine modeled on LevelDB: a skip-list memtable for O(log n) insertion and point lookup, a write-ahead log appending every operation before the memtable is modified, frozen memtables flushed to immutable SSTables, levelled compaction, Bloom filters to cut reads that touch disk for keys that are not there, and tombstone deletion reclaimed during compaction. Exposed over a ctypes FFI to Python/FastAPI and a React dashboard.

nexus_cluster · Distributed Consensus Engine | *Python, Raft, TCP*

- Implemented Raft from the paper with no consensus library: leader election, log replication and term management over a custom TCP transport, durable term and vote so a restarted node cannot vote twice, log compaction by snapshot, and a read-index path for linearizable reads that do not append an entry. A simulation harness checks the paper's safety properties after every step as nodes fail and rejoin.

nano_match · Limit Order Book and Matching Engine | *C++*

- Built a price-time priority matching engine covering the full order lifecycle. Eliminated all heap allocation after startup with a custom pool, reaching 12M requests/s at a 50ns median on one core.

Additional: custom_mem_alloc (thread-safe `malloc` over one `mmap` region, benchmarked against glibc) · nexus_editor (CRDT editor, Go) · photon_tracer (CPU path tracer, 1080p at 50 spp in half a second on 32 threads) · neon_vm (bytecode VM) · cpu_rasterizer and rasterizer_engine (software 3D pipelines, JS and C++/SDL2) · radiant_slice (FPS systems, Unreal Engine 5) · thermal_monitor, terminal_dashboard, valo_scout, cal-cli. 30 public repositories, 11 with live demos.

ACTIVITIES

Mixed Martial Arts: two-time national champion, 2020 to 2024, training six days a week alongside full-time coursework.